

Color recognition

1. First determine whether the version of the image transmission module is the AI version firmware. **If you already know the firmware version, you can ignore this step** and use the **wifi_ver** command to query, as shown in the figure below

```
[2024-08-06 14:53:29.948]# SEND ASCII>
wifi_ver
[2024-08-06 14:53:30.010]# RECV ASCII>
YAHBOOM VerSion:AI_V0.5.0
```

2. Switch the mode to the color recognition mode. Use the following command to send **ai_mode:3** through the serial assistant. After successful configuration, the module will restart and print the current firmware version. You can start color recognition, as shown in the figure

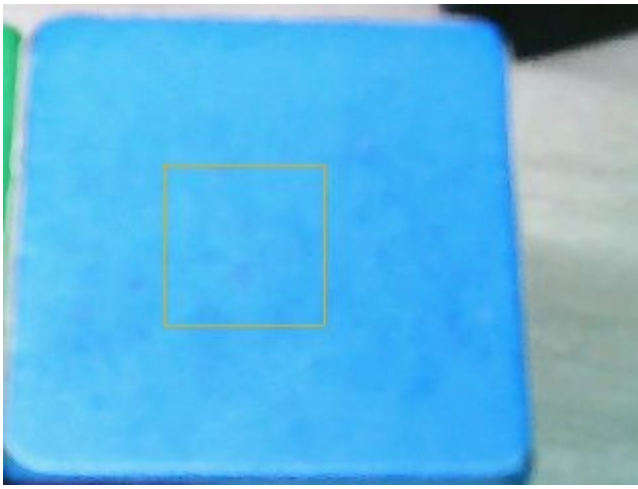
```
[2024-08-06 15:03:04.205]# SEND ASCII>
ai_mode:3
[2024-08-06 15:03:04.341]# RECV ASCII>
ESP-ROM:esp32s3-20210327
Build:Mar 27 2021
rst:0x3 (RTC_SW_SYS_RST),boot:0x8 (SPI_FAST_FLASH_BOOT)
Saved PC:0x40375a0c
SPIWP:0xee
mode:DIO, clock div:1
load:0x3fce3818,len:0x514
load:0x403c9700,len:0x4
load:0x403c9704,len:0xb58
load:0x403cc700,len:0x2a8c
entry 0x403c98fc

[2024-08-06 15:03:04.886]# RECV ASCII>
[0:32mI (549) cpu_start: Multicore app [0m
[0:32mI (549) esp_psram: Found 2MB PSRAM device [0m
[0:32mI (550) esp_psram: Speed: 80MHz [0m
[0:32mI (550) cpu_start: Pro cpu up. [0m
[0:32mI (554) cpu_start: Starting app cpu, €
[2024-08-06 15:03:07.070]# RECV ASCII>
YAHBOOM AI VerSion:AI_V0.5.0
```

Note: If the current mode is already the color recognition mode, the module only returns **ok**

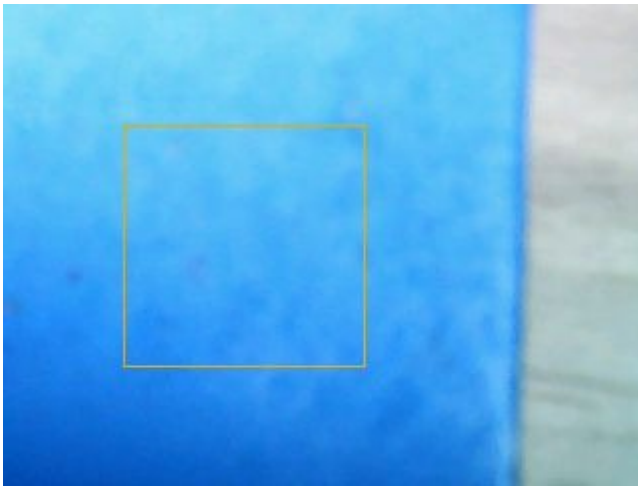
```
[2024-08-06 15:05:32.444]# SEND ASCII>
ai_mode:3
[2024-08-06 15:05:32.503]# RECV ASCII>
OK
```

3. To start recording the color, you need to send **KEY_MENU** through the serial assistant and wait for the recording box to appear on the screen

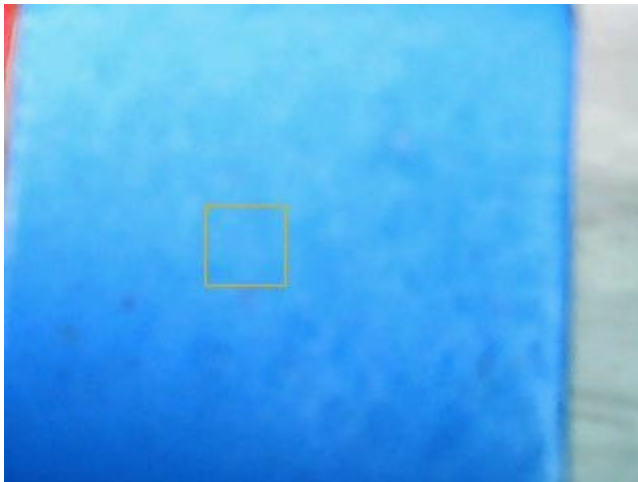


Warm Tips: You can send **KEY_UPUP**** through the serial port to increase the recording box selection, and send **KEY_DOWN** to reduce the box selection**

KEY_UPUP:



KEY_DOWN:

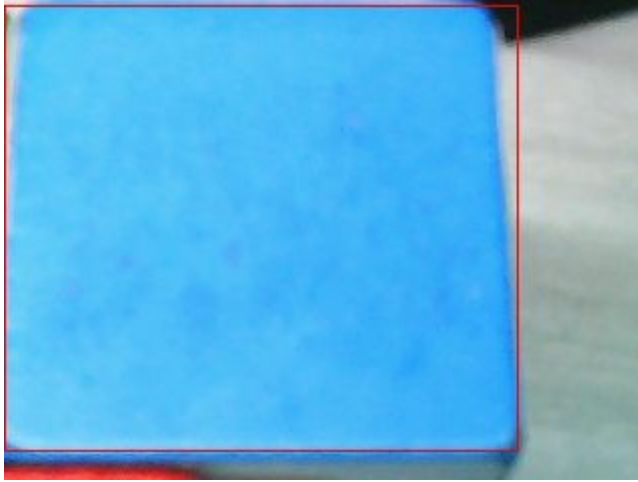


4. After selecting the color, send **KEY_PLAY** through the serial assistant. After confirming the selected color, the recording box on the screen will disappear, and then the color just recorded will be directly recognized. And send the coordinates of the upper left corner and lower right corner of the first recognized box through the serial port.

The specific content is **\$upper left corner x-axis coordinate, upper left corner y-axis coordinate, lower right corner x-axis coordinate, lower right corner y-axis coordinate, #**

Note: You need to open the screen to start recognition. If the screen is closed in the middle, the recognition will also be closed.

Result:



Serial port information:

```
KEY_MENU OK
KEY_PLAY OK
$0, 3, 260, 231, # $0, 3, 256, 228, # $0, 3, 260, 231, # $0, 3, 256, 231, # $0, 3, 256, 231, #
$0, 3, 260, 228, # $0, 3, 256, 228, # $0, 3, 256, 231, # $0, 3, 260, 228, # $0, 3, 260, 231, #
$0, 0, 256, 228, # $0, 0, 260, 228, # $0, 0, 260, 228, # $0, 0, 260, 228, # $0, 0, 256, 228, #
$0, 0, 256, 228, # $0, 3, 256, 228, # $0, 3, 260, 228, # $0, 3, 256, 228, # $0, 3, 260, 228, #
$0, 3, 256, 228, # $0, 3, 252, 228, # $0, 0, 256, 228, # $0, 0, 256, 225, # $0, 0, 256, 228, #
$0, 3, 248, 228, # $0, 3, 252, 228, # $0, 3, 252, 225, # $0, 3, 252, 228, # $0, 3, 252, 228, #
$0, 0, 252, 225, # $0, 0, 248, 225, # $0, 0, 252, 225, # $0, 0, 252, 222, # $0, 0, 256, 225, #
$0, 0, 252, 222, # $0, 0, 256, 225, # $0, 0, 252, 225, # $0, 0, 252, 222, # $0, 0, 256, 225, #
$0, 0, 252, 225, # $0, 0, 252, 222, # $0, 0, 252, 225, # $0, 0, 252, 225, # $0, 0, 252, 225, #
$0, 0, 256, 222, # $0, 0, 252, 225, # $0, 0, 252, 222, # $0, 0, 252, 225, # $0, 0, 252, 222, #
$0, 0, 252, 222, # $0, 0, 248, 219, # $0, 0, 248, 222, # $0, 0, 252, 222, # $0, 0, 252, 219, #
$0, 0, 252, 222, # $0, 0, 244, 219, # $0, 0, 248, 222, # $0, 0, 248, 222, # $0, 0, 244, 222, #
```

5. When you want to add recognized colors, you can repeat steps 3 and 4. When you want to delete recognized colors (starting from the last color), the mode is in the detection state and then send **KEY_PLAY** from the serial assistant to delete it.

